



Hypersonic boost-glide systems: Flight mechanics and plasma parameters evaluation through aero-thermo-chemical computational fluid dynamics

Ragnoli Giona^{a,*}, Savino Raffaele^b, Cecere Anselmo^b, Rigamonti Marco^c

^a Italian Air Force Academy, Pozzuoli, Italy

^b University Federico II, P.le Vincenzo Tecchio 80, Naples, 80125, Italy

^c Aero-Space Engineering Group, Italian Air Force Flight Test Department, Italy

ARTICLE INFO

Communicated by Cummings Russell

Keywords:

Hypersonic boost-glide systems
Flight mechanics simulation
Non-equilibrium CFD
Chemical species ionization
Collision frequency
Plasma frequency

ABSTRACT

Currently, the conception, design, and development of hypersonic space-borne systems is a central topic of discussion among researchers and industrial stakeholders. In the design phase of a hypersonic manoeuvrable vehicle, it is essential to characterise the thermal heating and manage the problem of the communication blackout, in addition to other challenges concerning guidance, navigation and control, and propulsion. The interference of electromagnetic waves propagating in plasma affects the datalink and the radar detection of hypersonic targets. This work aims to provide valuable information about the aero-thermo-chemical behaviour of the fluid in the region around a hypersonic glider. Launch and re-entry phases of boost-glide systems have been modelled to perform trajectory predictions and assess range and flight paths under the effect of different manoeuvres, defining their possible performances. The resulting flight conditions are used to study the flow field surrounding a glider geometry similar to the Hypersonic Technology Vehicle-2. The thermodynamic and chemical fields around the body have been determined, and plasma parameters for novel electromagnetic interaction studies have been evaluated. Computational fluid dynamics simulations have been carried out to solve the averaged Navier-Stokes equations with the $k - \omega$ model for turbulent flows, coupled with the energy equation. Chemical non-equilibrium phenomena involving electronic energy relaxation as well as ionisation reactions have also been included. A gas mixture potentially composed of 11 species has been considered, namely oxygen and nitrogen derivatives. The electron number density is computed to evaluate the electron plasma and collision frequencies, essential parameters to characterise the electromagnetic interference caused by the plasma sheath. Among other results, high temperatures are reached at the nose region of the body in specific conditions of velocity and altitude, and non-negligible dissociation or ionisation of chemical species occurs. Plasma frequencies up to tens of GHz are reached in a region close to the body surface, while lower interference characterises the extended ionised wake.

1. Introduction

Decades of research are offering continuous advancements and opportunities in high-atmospheric flight [1]. As flight speeds and associated energy levels increase, different physical phenomena become important; hence, the term hypersonic was introduced to refer to speeds where important energy exchange mechanisms occur in the fluid. Due to the excitation of vibration and electronic energy levels, chemical reactions and the eventual ionisation of the gas surrounding the flying object are no longer negligible. According to [2], the hypersonic regime is the range of speeds where aerodynamic heating dominates

the physics of flows and is frequently associated with objects moving with a Mach number greater than 5. The conception, design, and development of hypersonic space-borne systems is a central topic of discussion among researchers and industrial national stakeholders, and it involves a variety of fields such as aerodynamics, material technology science, propulsion, guidance, navigation and control (GNC), and communications engineering [3–5]. The hypersonic capability could potentially transform high-atmospheric travel and provide easy access to space [6].

As space systems and civilian hypersonic applications are being developed, it is also true that space-faring nations are investing significant

* Corresponding author.

E-mail addresses: g.ragnoli@studenti.unina.it (G. Ragnoli), raffaele.savino@unina.it (R. Savino).

<https://doi.org/10.1016/j.ast.2024.109092>

Received 3 January 2024; Received in revised form 15 March 2024; Accepted 22 March 2024

Available online 29 March 2024

1270-9638/© 2024 Elsevier Masson SAS. All rights reserved.

Nomenclature

Abbreviations

2D	Two-dimensional
3D	Three-dimensional
AFB	Air Force Base
CFD	Computational Fluid Dynamics
CG	Centre of gravity
EM	Electro-magnetic
HCM	Hypersonic Cruise Missile
HGV	Hypersonic Glide Vehicle
HTV	Hypersonic Technology Vehicle
<i>l.e.</i>	Leading edge
RV	Re-entry vehicle
<i>s.l.</i>	Sea level
<i>vac</i>	Vacuum

Physics Constants

ϵ_0	Vacuum permittivity.....	8.854×10^{-12}	Fm^{-1}
κ	Boltzmann constant.....	1.381×10^{-23}	JK^{-1}
e	Electron charge.....	1.602×10^{-19}	C
g	Gravitational acceleration.....	9.81	$\text{m}\cdot\text{s}^{-2}$
m_e	Electron mass.....	9.109×10^{-31}	kg
r_e	Earth radius.....	9.371×10^6	m

Symbols

α	Angle of attack.....	rad
β	Side-slip angle.....	rad
δ	Flight path heading angle.....	rad
γ	Flight path elevation angle.....	rad
λ_e	Mean free-path of electrons.....	m
μ	Thrust vector angle.....	rad
μ_e	Electrons dynamic viscosity.....	$\text{kg}\cdot\text{m}^{-1}\text{s}^{-1}$
ν	Bank angle.....	rad
ν_c	Total collision frequency.....	s^{-1}
ν_{es}	Electrons-species collision frequency.....	s^{-1}
ω_p	Plasma pulse.....	$\text{rad}\cdot\text{s}^{-1}$
ϕ	Re-entry velocity incidence.....	rad
ρ	Air density.....	$\text{kg}\cdot\text{m}^{-3}$

ρ_s	Density of species s.....	$\text{kg}\cdot\text{m}^{-3}$
T	Temperature.....	K
a	Inertial acceleration.....	$\text{m}\cdot\text{s}^{-2}$
B	Ballistic coefficient.....	$\text{kg}\cdot\text{m}^{-2}$
C_D	Drag coefficient	
C_L	Lift coefficient	
D	Drag force.....	N
D_e	Electrons diffusion coefficient.....	$\text{m}^2\cdot\text{s}^{-1}$
f_p	Plasma frequency.....	s^{-1}
h	Altitude.....	m
L	Lift force.....	N
L/D	Lift-to-drag ratio	
M	Mach number	
m	Payload mass.....	kg
m_s	Mass of species s.....	kg
n	Degrees of freedom of molecules	
N_s	Number of species s	
n_s	Number-density of species s.....	m^{-3}
R	Pull-up radius parameter.....	m
r_n	Radius of the neutral species n.....	m
S	Reference surface.....	m^2
t	Time.....	s
V	True airspeed.....	$\text{m}\cdot\text{s}^{-1}$
W	Weight force.....	N
Z	Number of lacking charges of the ion	
T	Thrust force.....	N

Reference frame

τ_A	Aerodynamic reference frame
τ_B	Body reference frame
τ_E	NED reference frame
τ_W	Wind reference frame
lla	Latitude, longitude and altitude
NED	North-East-Down reference frame
$WGS-84$	World Geodetic System 1984
x, y, z	Local horizontal reference frame axis
x_E, y_E, z_E	NED reference frame axis

amounts of money in hypersonic military capabilities to gain an edge in strategic situations [7]. For this reason, they are also military powers assessing the capabilities of hypersonic cruise missiles, boost-glide systems, interceptor missiles, reusable spacecraft, space launch vehicles, and gun-launched projectiles [8]. According to the Congressional Research Service [9], hypersonic strike weapons are in development for applications up to intercontinental ranges, and they are believed to bring weapons deterrence to the next level. They combine the abilities to perform prolonged flight at extremely high speeds, in the order of kilometres per second, and to manoeuvre during the flight.

There are two main technologies of hypersonic missiles: hypersonic boost-glide systems and hypersonic cruise missiles (HCMs). The former consists of a ballistic rocket booster and a hypersonic glide vehicle (HGV) and the latter is equipped with an air-breathing engine (e.g., a ramjet or a more powerful scramjet). HGVs rely on an elevated lift-to-drag ratio to sustain flight within the atmosphere while descending at high speed, exploiting their potential energy gained after launch. The peak speed attained after the boost phase and the glide vehicle's lift-to-drag ratio are the main factors influencing the potential operative distance. Even if the majority of the flight paths undertaken by HGVs occur inside the atmosphere, depending on the trajectory of the ballistic missile booster carrying the HGV, a portion of it may also be exo-atmospheric. Multiple trajectories exist that can augment range, sustain speed, and reduce the thermal strain. A peculiar trajectory is known as

the “skip-glide” trajectory, where the glider bounces on the upper layers of the atmosphere [10]. Hence, the study of re-entering gliders is essential to developing sensors, systems, technologies, or more simply, new approaches to detect and track hypersonic systems, and to take action against their threatening use. At the same time, it is essential to characterise both flight mechanics and fluid fields around high-speed vehicles to manage the challenges of high-atmospheric flight and develop hypersonic capability. The Italian Air Force plays a key role in defining the future of research and development programmes to guarantee human welfare and progress and to improve National Defence against entirely new classes of offensive hypersonic technologies. For this reason, the study was conducted in collaboration with the Aero-Space Engineering Group, which aims to develop new capabilities to expand the boundaries of the traditional aeronautical domain into space.

1.1. Plasma sheath and impact on communications

While descending into the atmosphere of the Earth, any hypersonic vehicle experiences extreme thermal heating. As a result of the high pressure and heat generated, the air molecules around the vehicle dissociate and lose their electrons, creating a layer of plasma adjacent to the body [11]. Hence, the plasma that surrounds hypersonic vehicles is a high-energy ionised gas consisting of positively charged ions and negatively charged electrons generated by extreme temperatures. This

plasma layer, also called sheath, has several unique properties that differentiate it from a normal gas. For one, the plasma is an excellent conductor of electricity. Additionally, the plasma can absorb and reflect electromagnetic radiation. The high electrical conductivity of the plasma can interfere with the communication systems and onboard electronics, both necessary for precision in manoeuvres. The plasma sheath also absorbs and reflects radio waves in communication systems, causing signal loss and distortion [12].

To overcome the disturbance of telecommunications, known as communication blackout, researchers are developing innovative transmitters working at high frequencies. Moreover, plasma interference could significantly impact the accuracy of radar measurements commonly used to perform the detection of re-entering objects as well as for air defence purposes. The plasma sheath can interfere with radar signals, causing false detection and a reduced detection range. For this reason, research is currently assessing advanced radar systems that can operate at high frequencies to penetrate the plasma sheath and accurately detect hypersonic vehicles. Hence, it is essential to characterise the thermofluid-dynamic and chemical fields around hypersonic vehicles to carry out specific studies of Doppler frequency shifts for radar detection. Some previous studies analysed the infrared radiation of HTV-2-type vehicles at different angles of attack [13], while the radar detection of hypersonic gliders remains quite an unexplored field, especially for complex geometries. Some examples show how to study electromagnetic (EM) interactions on simple blunt shapes, but the plasma parameters used come from old experimental databases and are not explained in detail [14–16].

CFD offers the possibility to perform detailed analyses of complex hypersonic flows with shock-boundary layer interaction to determine the ionisation of chemical species and the effect on plasma parameters. These parameters may be used to conduct electromagnetic interaction research on hypersonic gliders. To sum up, this research aims to characterise the variation of some useful plasma parameters at different flight conditions, obtained using a made-on-purpose numerical trajectory simulation tool, which is also used to evaluate the effect of different bank manoeuvres on the final range.

1.2. Mission under study

To characterise the thermo-fluid-dynamic and chemical fields surrounding a HGV, the real mission of Hypersonic Technology Vehicle 2 (HTV-2) is taken as a reference. According to the mission's environmental assessment [17], the HTV-2 flight test was meant to show how aerodynamics, guidance and control theories, high-temperature materials, and thermal protection systems work for prolonged hypersonic flight in the upper atmosphere. The intent was to conduct two flight tests to support the development and demonstration of hypersonic technologies. The mission plan is shown in Fig. 1. The first flight occurred in 2010, following the A route. The test was cut short when the vehicle unexpectedly started rotating out of control. Consequently, the A route was also used for the second flight test in 2011. The mission can be summarised in the following steps:

- 1) *Launch and ascent*: the booster lifts off from Vandenberg AFB. All three spent motor stages splash down at various points in the ocean, between 100 and 2,000 nautical miles off the coast of California;
- 2) *Fairing and payload separation*: the jettison of the fairing occurs in the first part of the ascent while the glider separates outside the atmosphere;
- 3) *Payload descent*: the glider uses autonomous flight control to manoeuvre during the hypersonic glide at an altitude between 45 and 76 km above the ground, and to follow the intended path;
- 4) *Impact*: approaching the terminal end of its flight, at an altitude of approximately 30.5 km, the glider performs a dive manoeuvre towards the designated ocean impact area, USAKA/RTS, about 7700 km away from the launch pad.

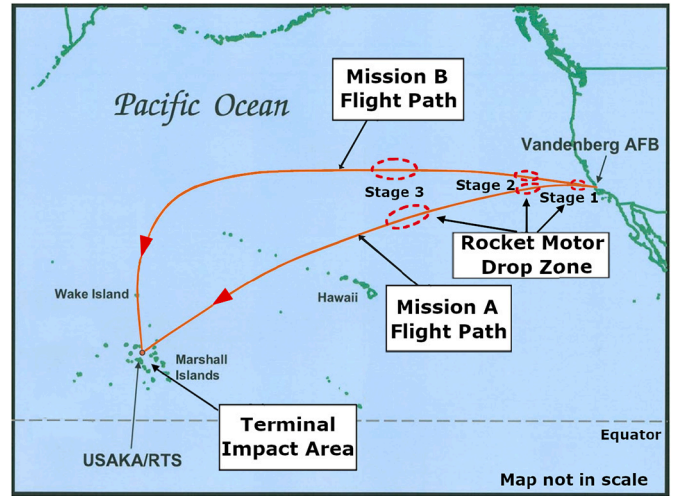


Fig. 1. HTV-2 planned routes (A and B) departing both from Vandenberg AFB.

2. Flight mechanics and simulation setup

2.1. Preliminary assumptions and methods

The developed trajectory simulation tool aims to obtain flight parameters to be used as input in the CFD analysis. The model is implemented in MATLAB [18], solving the point-mass dynamics for each flight phase. More in detail, following the same approach of the previous study [19], the mission is divided into different phases:

- 1) *Launch phase*: Minotaur IV Lite launcher is used;
- 2) *Ballistic phase*: after the burnout of the third stage, a ballistic trajectory is followed, the payload separates from the launch vehicle during this phase;
- 3) *Direct re-entry phase*: the drag force increases due to higher density and angle of attack. The vehicle slows down without producing lift force;
- 4) *Pull-up phase*: the vehicle slowly rotates towards a lower angle of attack, this manoeuvre aims to generate high-lift, low-drag orientation;
- 5) *Equilibrium glide phase*: the orientation at the end of the pull-up phase allows a higher lift-to-drag ratio, and the glider is capable of maintaining levelled flight;
- 6) *Impact phase*: once the vehicle is close to the target, a sinking manoeuvre is performed to reach it at high speed.

The launch phase is modelled following reference [20], which also offers estimations for additional avionics and structural mass, 450 kg and 800 kg respectively. The characteristics of Minotaur IV Lite booster are reported in Table 1.

Aerodynamic coefficients for each stage are provided assuming suitable values for a streamlined body, similar to other cylindrical-shaped rockets during the ascent. Lift and drag forces are neglected above 80 km altitude, where the Knudsen number is below 0.001 and the atmosphere can not be considered as a continuum regime. Starting from the initial condition, the state at the successive time step is evaluated assuming constant acceleration in the time interval, resulting from point mass dynamics in equations (1). The procedure is iterated until the burnout of each stage. In addition, the angle between the thrust force and the velocity vector of each booster stage is adjusted to reach the expected value of the flight path angle, which provides a splash-down point in the mission-planned area. So, the inclination of the thrust vector is varied proportionally to the difference between the instantaneous flight path angle and the desired one. Thrust and specific impulse parameters are assumed constant for each stage.

Table 1

Minotaur IV Lite technical data of performances and characteristics of each solid stage.

Stage	Diameter (m)	Length (m)	Propellant (kg)	Total mass (kg)	Specific impulse (s)	Thrust (kg)	Burn time (s)
1st	2.3	8.4	45,370	48,990	259 (s.l.) 282 (vac)	209,000 (s.l.) 226,800 (vac)	56.4
2nd	2.3	6.0	24,490	27,670	309 (vac)	124,700 (vac)	60.7
3rd	2.3	3.3	7,070	7,710	300 (vac)	29,480 (vac)	72

After the burnout of the third stage, the object starts a ballistic phase, which is considered to last at the pierce point. During this time interval, the payload separates from the rocket and begins its re-entry flight until the beginning of the pull-up manoeuvre. The pull-up phase implements a basic model of the trajectory presented in reference [19], assuming no oscillations during the manoeuvre. Both lift-to-drag ratio, L/D , and ballistic coefficient, B_L , are assumed as constants during the glide phase, which is interrupted at an altitude of 30.5 km. While the model presented by Acton uses an analytical solution for the equations of the glide phase, in this study the flight trajectory is numerically solved by integrating equations (12) with a Runge-Kutta fourth-order method. In each flight phase, the point-mass dynamics are solved considering the equilibrium of forces acting on the body. Equations are solved in a local North-East-Down (NED) reference frame and the resulting position is converted into latitude-longitude-altitude coordinates (lla). All flight parameters and aerodynamic coefficients are obtained. The assumptions made to build the model are:

- 1) US Standard Atmosphere 1976 is used;
- 2) The trajectory is planar until the beginning of the gliding;
- 3) The launch direction defines the great circle arc where downrange is computed;
- 4) The Earth is considered as a WGS-84 ellipsoid;
- 5) The rotation of the Earth is neglected during the entire flight [21]. As a consequence, the Coriolis effect has not been taken into account in the equations;
- 6) The centrifugal force of the glider flying over a non-flat Earth is neglected in the glide phase;
- 7) The gravitational acceleration is considered as constant, $g = 9.81 \text{ m/s}^2$;
- 8) The bank angle is assigned and constant during turnings;
- 9) The equations are solved in the local North-East-Down (NED) reference frame. To avoid excessive approximations in downrange and cross-range computation, the NED is refreshed to the ground-projected position of the glider every time the distance from the origin of the previous reference frame is greater than 10 km.

2.2. Detailed modelling and equations

2.2.1. Launch phase

Fig. 2 shows that the dynamics of the body depend on the lift, L , drag, D , weight, W , and thrust, T , forces. The flight path angle, γ , is defined as referring to the local horizontal direction, x , positive when the velocity vector, V , is above it.

While the velocity vector is always considered aligned with the longitudinal axis of the body, the thrust is tilted with an angle μ , which is varied to perform attitude control. The equilibrium equations are shown below:

$$\begin{cases} ma_x = T \cos(\gamma - \mu) - D \cos \gamma - L \sin \gamma \\ ma_z = -T \sin(\gamma - \mu) + D \sin \gamma - L \cos \gamma + mg - m \frac{(V \cos \gamma)^2}{r_e} \end{cases} \quad (1)$$

where $m = 1000 \text{ kg}$ is the mass of the glider, a is the acceleration, r_e is the radius of the Earth used to compute the centrifugal force. The lift force ($L = \frac{1}{2} \rho V^2 S C_L$) and the drag force ($D = \frac{1}{2} \rho V^2 S C_D$) are computed, where ρ is the air density, and S is the reference surface, which

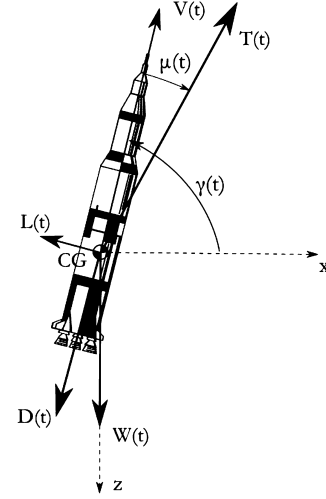


Fig. 2. Force schematic of the launch vehicle. The angles μ and γ are shown: for the launch phase simulation, the thrust vector is tilted proportionally to the difference between the instantaneous flight path angle and the desired one.

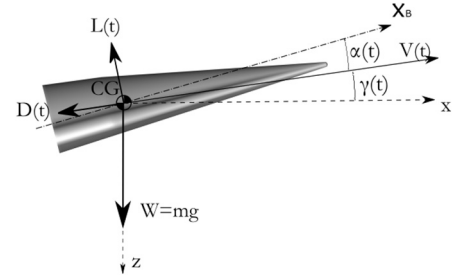


Fig. 3. HTV-2 glider scheme of forces (lift, L , and drag, D , are null during the entire ballistic phase, above the pierce point).

is assumed to be equal to the section area of the booster. The lift coefficient, C_L , is zero for the launch phase, while the drag coefficient, C_D , is set to 0.5 for the first and second stages. At altitudes above 80 km, the Knudsen number (computed as the ratio between mean molecular distance and length of the body in Fig. 5) is above a value of 0.001, and the continuum regime hypothesis is not valid. Due to the very high rarefaction, the atmospheric effects are neglected above 80 km since they are several orders of magnitude lower than the thrust force.

2.2.2. Ballistic and re-entry phases

After the burnout of the third stage, which happens at an altitude of nearly 130 km, a ballistic trajectory is followed for 81 seconds. At the apogee of the trajectory (around 147 km), the payload separates from the final stage (270 seconds after launch). The dynamics of these phases are obtained from equations (1), with null thrust and neglecting lift and drag until the pierce point (about 96 km), where the re-entry phase starts. During the re-entry, C_D is evaluated using the Newtonian theory for rarefied flows. The mass, m , is the sum of the empty booster and payload masses until separation. Subsequently, it becomes equal to the mass of the payload (see Fig. 3).

As altitude decreases, due to the swift augmentation of atmospheric density, a re-entry vehicle (RV) that approaches the atmosphere at a slight angle may encounter a boost in its lift force. This results in a phugoid trajectory, where the RV bounces along the upper layers of the atmosphere. To prevent a glider from skipping off the atmosphere during re-entry, one strategy is to use a flat attitude in such a way as to expose the lower surface to the atmosphere, reducing speed with a low-lift configuration ($C_L = 0$). During this phase, which starts at the pierce point, the ballistic coefficient defined in equation (2) is assumed to be constant.

$$B = \frac{m}{C_D S} \quad (2)$$

In the previous study [19], a value of $B_s = 7 \text{ kg/m}^2$ is considered, which is an order-of-magnitude smaller than expected value; in this paper, the value has been derived by the application of the Newtonian theory, with the following formula (also presented in [19]):

$$B_s = \frac{m}{2S \sin^3 \phi} \quad (3)$$

where ϕ is the angle between the normal to the surface and the velocity vector. Considering a flat re-entry ($\phi = 90^\circ$) and a reference surface $S = 4.7 \text{ m}^2$, the assumed ballistic coefficient is $B_s = 100 \text{ kg/m}^2$. This different value has a very low effect on final parameters at the end of the direct re-entry phase because the phase lasts just twelve seconds.

2.3. Pull-up phase

During this phase, a manoeuvre is performed to reduce the angle between the longitudinal axis of the glider and the velocity vector to obtain a low-drag, high-lift configuration suitable for gliding. If the rotation is swift, the glider does not make a smooth transition into equilibrium gliding but follows an oscillating trajectory. In fact, due to the constant lift coefficient assumption, the increasing lift force during the descent causes the glider to overshoot its equilibrium point. Noticing that the flight path angle is defined as positive in the opposite direction, expressions in reference [19] become:

$$\gamma - \gamma_2 = \frac{L}{D} \left(1 + \frac{R}{r_e} - \frac{gR}{V_3^2} \right) \frac{V_3 - V}{V_3} \quad (4)$$

In the previous and following equations, the numerical indexes refer to the endpoints of each phase as they are presented in Section 2.1. It is possible to isolate the value of R which guarantees a flight path angle equal to zero at the end of the manoeuvre, $\gamma = \gamma_4 = 0$:

$$R = \frac{\left(1 - \gamma_2 \frac{D}{L} \frac{V_3}{V_3 - V_4} \right)}{\frac{1}{r_e} - \frac{g}{V_3^2}} \quad (5)$$

and substitute this expression into equation (6).

$$h_3 - h = R \frac{L}{D} \left[\gamma_2 \left(\frac{V - V_3}{V_3} \right) - \frac{1}{2} \frac{L}{D} \left(1 + \frac{R}{r_e} - \frac{gR}{V_3^2} \right) \left(\frac{V_3 - V}{V_3} \right)^2 \right] \quad (6)$$

Once the final altitude is set, $h_4 = 47000 \text{ m}$, the final velocity is computed using equation (7).

$$V_4 = \frac{(h_3 - h_4) \left(\frac{1}{r_e} - \frac{g}{V_3^2} \right) \frac{D}{L} - \frac{1}{2} \frac{D}{L} \gamma_2^2 - \frac{1}{2} \gamma_2}{\frac{1}{2} \frac{(-\gamma_2)}{V_3}} \quad (7)$$

With equation (5), and the definition of the R parameter in expression (8), it is possible to obtain the ballistic coefficient of the pull-up phase, B_L . Using equation (2), C_D and C_L are obtained, considering $L/D = 2.6$.

$$R = \frac{2B_L}{\rho(h_4)} \frac{D}{L} \quad (8)$$

The duration of the manoeuvre is:

$$t_4 - t_3 = \left(1 - \frac{V_4}{V_3} \right) \frac{L}{D} \frac{R}{V_3} \quad (9)$$

It is finally possible to compute velocity, downrange, flight path angle, and altitude at each equal-time step, with equation (10), (11), (4), (6), respectively.

$$V = V_3 \left(1 - \frac{D}{L} \frac{V_3(t - t_3)}{R} \right) \quad (10)$$

$$x - x_3 = V_3 \left[(t - t_3) - \frac{D}{L} \frac{V_3(t - t_3)^2}{2R} \right] \quad (11)$$

2.3.1. Glide phase

At the end of the pull-up phase, equilibrium gliding is established with a ballistic coefficient B_L . The lift-to-drag ratio, L/D is assumed constant. It is possible to define transformations between reference frames, to pass from the local NED (indicated with subscript E) to the body reference frame (indicated with subscript B), passing through the wind reference frame and the aerodynamic reference frame.

$$\mathcal{T}_E \xrightarrow{\delta, \gamma} \mathcal{T}_W \xrightarrow{\nu, -\beta} \mathcal{T}_A \xrightarrow{\alpha_B} \mathcal{T}_B$$

With reference to Fig. 4, δ is the flight path heading angle, γ is the flight path elevation, ν is the bank angle, β is the side-slip angle, and α is the angle of attack.

To build the model, it is assumed that the side-slip angle, β , is constantly equal to zero, and the body reference frame is aligned with the aerodynamic one, $\alpha \sim 0$. With these assumptions, the equations of motion [22] become:

$$\begin{cases} \dot{V} = \frac{T - D}{m} - g \sin \gamma \\ \dot{\gamma} = \frac{1}{mV} (L \cos \nu - mg \cos \gamma) \\ \dot{\delta} = \frac{L \sin \nu}{mV \cos \gamma} \end{cases} \quad (12)$$

Equations (12) represent the translational dynamics of a vehicle that maintains a constant zero side-slip angle and where the thrust is always aligned with the velocity vector. In the implemented model, thrust is null since the body is a glider. The inclination of the lift vector, due to the bank angle, results in the variation of the flight path heading angle, δ . Additional kinematic equations, which represent the centre of gravity translation, are required to solve the dynamics.

$$\begin{cases} \dot{x}_{E,G} = V \cos \gamma \cos \delta \\ \dot{y}_{E,G} = V \cos \gamma \sin \delta \\ \dot{h} = V \sin \gamma \end{cases} \quad (13)$$

where V is the airspeed, which, in general, is related to the wind. Due to the flight altitude, no wind is considered to compute the velocity.

$$V^2 = (V \cos \gamma \cos \delta)^2 + (V \cos \gamma \sin \delta)^2 + (V \sin \gamma)^2 \quad (14)$$

2.4. Flight mechanics validation

Tables 2 and 3 show a comparison of the simulation results with the previous models of D. Wright and J.M. Acton [19,20], which assumed that the analytical trajectory parameters were independent of each other. Values of $L/D = 2.6$, according to [19], and $h_4 = 47000 \text{ km}$ are assumed for the pull-up manoeuvre.

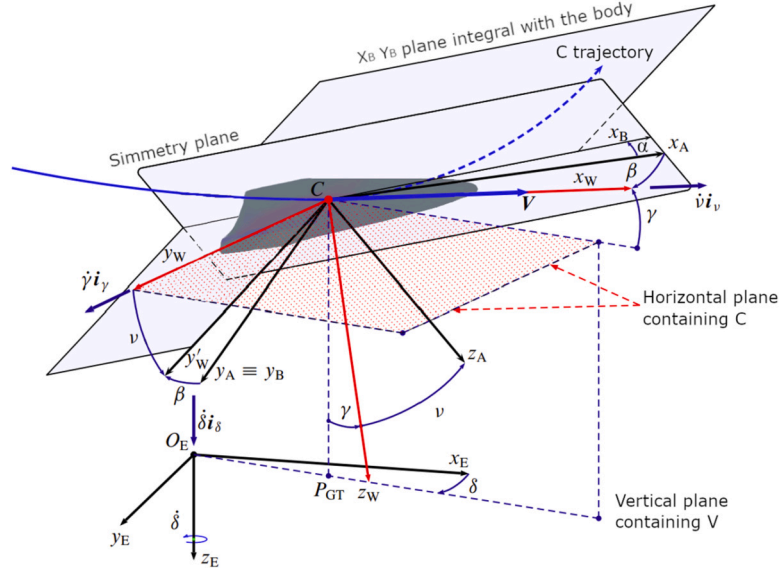


Fig. 4. Body (x_B, y_B, z_B) and NED (x_E, y_E, z_E) reference frames for the glide phase.

Table 2

Comparison of the launch phase simulation results with the model by D. Wright [20] to highlight the validity of the implemented model.

Parameter	D. Wright	Implemented model
Stage 1		
burnout time (s)	56.4	56.4
path angle (deg)	87	86.98
burnout altitude (km)	32	32.5
burnout speed (km/s)	1.4	1.38
burnout range (km)	n.a.	1.7
Stage 2		
burnout time (s)	60.7	60.7
path angle (deg)	10	9.96
burnout altitude (km)	~ 98	101
burnout speed (km/s)	3	3.09
burnout range (km)	~ 90	82
Stage 3		
burnout time (s)	72	71.9
path angle (deg)	3.7	3.7
burnout altitude (km)	127	130
burnout speed (km/s)	6.0	6.013
burnout range (km)	~ 380	384
Pierce point		
time (s)	435	435
altitude (km)	100	96
range (km)	1800	1930

Table 3

Comparison of the simulation results with the model by J.M. Acton [19]. Analytical relations among all parameters cause slight differences in the results.

Parameter	J.M. Acton	Implemented model
Pierce point		
altitude (km)	100	96
mission time (s)	435	435
path angle (deg)	- 5.03	- 6.3
velocity (km/s)	6.010	6.037
Entry (final)		
duration (s)	12	12
path angle (deg)	- 5.03	- 6.78
velocity (km/s)	n.a.	6.037
altitude (km)	n.a.	87.9
ballistic coefficient (kg/m ²)	7	100
Pull-up (final)		
altitude (km)	47	47
mission time (s)	620	570
path angle (deg)	0	0
velocity (km/s)	5.5	5.6
manoeuvre range (km)	1140	715
ballistic coefficient (kg/m ²)	13000	7475
R (km)	4900	3842

In conclusion, the differences between the assumptions of the two models are here presented. Acton's model hypotheses are:

1. Lift-to-drag ratio, $\frac{L}{D}$, ballistic parameters B_s , B_L , and R are varied independently;
2. The pierce point is assumed to be exactly at 100 km altitude;
3. An analytical model is used to represent the direct re-entry phase;
4. B_s is far from a real value.

The presented model assumptions are:

1. Analytical relations between parameters are considered;
2. The pierce point is at 96 km;
3. The direct re-entry phase is solved numerically as an extension of the ballistic phase;
4. B_s is estimated using equation (3).

3. Fluid dynamic analysis

3.1. Reference body geometry

Numerical simulations of complex hypersonic flows are performed to characterise the thermofluid-dynamic and chemical fields around the hypersonic glider. HTV-2-type geometry is used in both two-dimensional and three-dimensional simulations. A lifting-body hypersonic glider shape is considered, with a sweptback leading edge designed to guarantee a high lift-to-drag ratio. This allows it to manoeuvre over long distances during a 1500-seconds glide phase in the atmosphere. The outer shell of the aircraft is made of high-temperature-resistant materials with a melting point of over 3000 K, making it ideal for withstanding extreme heat. The geometry is shown in Fig. 5, with a ball head diameter $r_{nose} = 0.034$ m, and a length, width, and height of 3.67 m, 2.20 m, and 0.88 m respectively [23]. Based on previous studies available in the literature [20], the mass of the glider is $m = 1000$ kg. The longitudinal profile is used to carry out two-dimensional simulations.

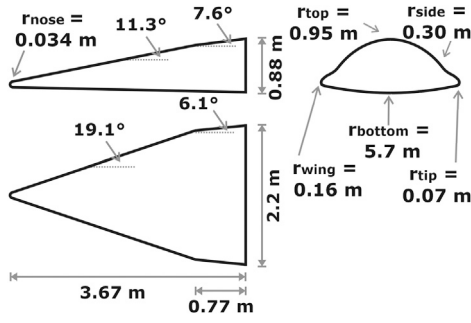


Fig. 5. Geometric features of the HTV-2 type body considered to perform CFD simulations.

Table 4

Chemical reactions considered in the numerical model.

Reaction	Third body, Y
$O_2 + Y \rightleftharpoons 2O + Y$	O_2, N_2, O, N, NO
$N_2 + Y \rightleftharpoons 2N + Y$	O_2, O, NO
$NO + Y \rightleftharpoons N + O + Y$	O_2, N_2, O, N, NO
$O^+ + e^- \rightleftharpoons O + e^- + e^-$	
$N^+ + e^- \rightleftharpoons N + e^- + e^-$	
$N_2 + N \rightleftharpoons N + N + N$	
$N_2 + O \rightleftharpoons NO + N$	
$NO + O \rightleftharpoons O_2 + N$	
$N + O \rightleftharpoons NO^+ + e^-$	
$O + O \rightleftharpoons O_2 + e^-$	
$N + N \rightleftharpoons N_2^+ + e^-$	

3.2. Aero-thermo-chemical model

The different flight conditions (in terms of altitude and Mach number) obtained with the trajectory simulation tool have been used to carry out CFD analyses in both two-dimensional and three-dimensional cases. It is well known that the temperature behind a normal shock layer, T , increases with the Mach number (M) and, if “high-temperature effects” are not considered, can be approximated with the stagnation temperature:

$$T_0 = T(1 + \frac{M^2}{n}) \quad (15)$$

where n is the number of degrees of freedom of the molecules. At the high speeds of hypersonic glide vehicles, feverish temperatures are reached in the fluid, and calorically perfect gases, as well as the degrees of freedom, cannot be considered anymore constant because several additional gas phenomena occur, including diffusive, turbulent and thermal non-equilibrium gas effects.

ANSYS Fluent [24] is used to solve the averaged Navier-Stokes equations with the $k-\omega$ model for turbulent flows, coupled with the energy equation. Since the test medium contains different species, transport equations are also solved along with the Navier-Stokes equations. Chemical non-equilibrium phenomena, involving electronic energy relaxation as well as ionisation reactions, have also been included using a code developed on the basis of the models of Dunn and Kang [25] and Evans et al. [26]. A gas mixture potentially composed of 11 species is considered, namely monoatomic oxygen and nitrogen, O and N , nitric oxide, NO , diatomic oxygen and nitrogen, O_2 and N_2 , the positive ions NO^+ , O^+ , N^+ , N_2^+ , O_2^+ , and electrons, e^- . Chemical reactions implemented in the numerical model are reported in Table 4.

The reaction rate of each chemical species (molecular, atomic or positively/negatively charged) is evaluated based on the law of mass action considering the non-equilibrium forward and backward rate given

in Table I of reference [25]. Transport coefficients for each species are computed as functions of the Lennard-Jones parameters using the kinetic theory. Electrons are treated as an additional gas species obeying the same governing equations valid for the others because under these conditions the flow can be considered a viscous flow in thermal equilibrium and chemical non-equilibrium. The transport properties are implemented with user-defined functions developed at the Department of Industrial Engineering of the University of Naples as follows [27,28]:

$$\mu_e = \frac{2.6693 \times 10^{-5} \pi \sqrt{5.4858 \times 10^{-4} T}}{10^{-0.8309923436 \times \log T \times 5.207708252 + 16}} \quad (16)$$

$$\lambda_e = 2.5 \times 7.88874 \times 10^7 \mu \quad (17)$$

$$D_e = \frac{\sum_{r=ions} D_r \rho_r m_e / m_r}{\sum_{r=ions} \rho_r} \quad (18)$$

The model has been validated with the experimental data of the NASA Langley Research Center RAM C-III, as shown in [29].

Temperature affects the translational, rotational, and vibrational degrees of freedom of molecules. This means that molecules tend to divide into atoms as their energy increases, forming atomic species and releasing electrons. In detail, oxygen molecules are the first to be split into atoms since their dissociation energy level is the lowest among other species in the air mixture. As temperature increases, nitrogen oxide is formed and, successively, nitrogen dissociates. Since dissociation reactions are endothermic, the energy is partially absorbed and specific heat coefficients are reduced. Due to these chemical reactions, electrons and ions are generated in hypersonic flow fields, and it is essential to obtain information about some plasma parameters involved in the electromagnetic characterisation of the ionised region [30]:

1. **Density of electrons:** the concentration of charged particles that results from the ionisation of air molecules grows with increasing speed and density. The electron density parameter in hypersonic plasma refers to the number of free electrons per unit volume, n_e . This parameter is important because it affects the behaviour of the plasma, including its electrical and thermal conductivity, in addition to the opacity of radiation.
2. **Plasma frequency:** it refers to the frequency at which the charged particles in the plasma oscillate collectively. In other words, it is the frequency at which the electric field of the plasma becomes significant enough to cause the charged particles to vibrate together. The plasma frequency is obtained from the plasma pulse, ω_p , given by the formula:

$$\omega_p = \sqrt{\frac{e^2}{\epsilon_0 m_e} n_e} \quad (19)$$

where n_e is the density of electrons, $e = 1.602 \times 10^{-19} C$ is the electron charge, and $m_e = 9.109 \times 10^{-31} kg$ is the mass of the electron, and $\epsilon_0 = 8.854 \times 10^{-12} \frac{F}{m}$ is the permittivity of free space. Hence, the plasma frequency, expressed in Hz , is given by equation (20).

$$f_p = \frac{\omega_p}{2\pi} \quad (20)$$

This parameter plays a crucial role in determining the behaviour of the plasma and its interaction with other waves. When the frequency of an external electromagnetic wave is close to the plasma frequency, the wave can be absorbed by the ionised layer, leading to additional ionisation and heating. More in detail, regarding the propagation of an incident electromagnetic wave with frequency f_w , when $f_w > f_p$, it could propagate through the plasma.

3. **Electron collision rate:** it refers to the rate at which electrons collide with other particles in the plasma. In hypersonic plasma, this collision frequency is affected by various factors such as the temperature and density of the plasma, the type of gas molecules in the plasma, and the energy state of the electrons. At high temperatures and densities, the electron collision frequency in hypersonic plasma is typically high. This can be attributed to the increased likelihood of collisions between electrons and other particles. In general, electrons with higher energy have a higher collision frequency than those with lower one. This is because higher energy electrons can overcome electrostatic repulsion from other particles more easily and are, therefore, more likely to collide. Moreover, the collision frequency is also affected by the type of gas molecules present in the plasma since larger particles increase the chances of collisions. In weakly-ionised plasma, the collision frequency, ν_c , is given by the interaction of electrons with other electrons, ν_{ee} , with ions, ν_{ei} , and with neutral molecules, ν_{en} .

$$\nu_c = \nu_{ee} + \nu_{ei} + \nu_{en} \quad (21)$$

According to [31], the interaction of two charges allows us to obtain the electrons-electrons collision term, ν_{ee} .

$$\nu_{ee} = \frac{2}{3\pi^{\frac{3}{2}} m_e^2} \frac{e^4}{\epsilon_0^2} n_e \left(\frac{4\kappa T_e}{m_e} \right)^{-\frac{3}{2}} \ln \Lambda = 2.567 \times 10^{-6} n_e T_e^{-\frac{3}{2}} \ln \Lambda \quad (22)$$

$$\Lambda \cong \frac{12\pi\epsilon_0\kappa T}{e^3} \left(\frac{\epsilon_0\kappa T}{n_e} \right)^{\frac{1}{2}} = 1.240 \times 10^7 T^{\frac{3}{2}} \left(\frac{1}{n_e} \right)^{\frac{1}{2}} \quad (23)$$

where $\kappa = 1.381 \times 10^{-23} \frac{m^2 kg}{s^2 K}$ is the Boltzmann's constant, T_e is the electron temperature expressed in Kelvin (approximated to the gas temperature, T , as for all the other chemical species in the mixture, $T_e = T_i = T_n = T$) and $\ln \Lambda$ is the Coulomb logarithm. In the same way, it is possible to obtain the electrons-ions collision term, ν_{ei} . Considering that the mass of the electron is negligible with respect to the mass of the ion, $m_e \ll m_i$, it is obtained:

$$\begin{aligned} \nu_{ei} &= \frac{1}{3\pi^{\frac{3}{2}} m_e^2} \frac{e^4 Z^2}{\epsilon_0^2} \left(\frac{2\kappa T}{m_e} \right)^{-\frac{3}{2}} \left(\sum_{i=1}^{N_i} n_i \right) \ln \Lambda \\ &= 3.634 \times 10^{-6} T^{-\frac{3}{2}} \left(\sum_{i=1}^{N_i} n_i \right) \ln \Lambda \end{aligned} \quad (24)$$

where Z is the number of lacking charges of the ions ($Z = 1$ in the considered gas mixture, for all ionic species), and n_i is the density of one of the N_i ionic species in the mixture. It is also possible to write the expression of the electrons-neutrals collision term, ν_{en} , considering $m_e \ll m_n$:

$$\nu_{en} = \frac{8\pi^{\frac{1}{2}}}{3} \left(\frac{2\kappa T}{m_e} \right)^{\frac{1}{2}} \sum_{n=1}^{N_n} r_n^2 n_n = 2.603 \times 10^4 \left(\sum_{n=1}^{N_n} r_n^2 n_n \right) T^{\frac{1}{2}} \quad (25)$$

where n_n is the density of one of the N_n neutral species in the mixture and r_n is the radius of the neutral particle.

3.3. Computational fluid dynamics setup

In this Section, information on the mesh used in the CFD simulations and numerical setup are presented. Moreover, a summary of the inlet and boundary conditions implemented for all simulations is given. In the numerical model for all the cases considered, the boundary of the external far field is defined as pressure far field, whereas pressure outlet is selected at the exit region as shown in Fig. 6. Structured surface mesh is used on the HTV-2 body surface, along with the extrusion

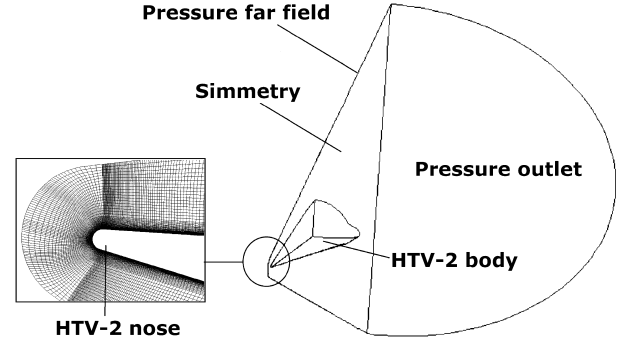


Fig. 6. Mesh of the numerical model and boundary conditions.

method with a fixed growth rate of 1.1 to create appropriate computational grids around the body. Thus, the number of layers is selected large enough to be able to use tetrahedral cells far from the vehicle. The mesh is composed of 1.3 million hexahedral cells with a first wall height of $4 \cdot 10^{-5} m$. This value has been chosen to maintain the value of the dimensionless boundary layer thickness parameter (y^+) below 30 in the worst case (i.e. the most critical condition corresponds to the simulation at 30 km altitude, Mach number equal to 18.6 and $Re = 2.56 \cdot 10^7$, as shown in Table 5, since high Reynolds numbers are associated with a thinner boundary layer). The far-field conditions have been evaluated using the US Standard Atmosphere 1976 model for each altitude case with a turbulent viscosity of 5% and a viscosity ratio equal to 10. The air composition is considered standard ($y_{O_2} = 0.22$, $y_{N_2} = 0.78$), while the HTV body surface is assumed to be a stationary non-catalytic surface. All the numerical analyses are carried out in the continuum regime and, so, a no-slip condition is considered. For the energy equation, a radiative thermal exchange is also included considering an emissivity coefficient $\epsilon = 0.85$ in a 300 K environment. Different flight conditions have been evaluated varying the Mach number (the flight conditions are summarised in Table 5). Two-dimensional studies have been carried out in all the cases, while three-dimensional analyses have been performed at an altitude $h = 40$ km, and a Mach number $M = 10.4$.

A constant angle of attack $\alpha = 15^\circ$ is considered in all the simulations. This value is a good compromise to take advantage of a relatively high lift-to-drag-ratio during the glide phase, without exposing the windward surface to critical heat loads [32]. The electron number density, n_e , is computed to evaluate the plasma frequency using the formula (20). A steady-state solution is selected. A density-based implicit coupled solver is chosen to solve the governing equations of continuity, momentum, and energy simultaneously. The Advection Upstream Splitting Method (AUSM) is used for spatial discretisation of inviscid fluxes, whereas diffusion terms are discretised by a first-order central differencing scheme. Temporal terms are also discretised through a first-order scheme. The discretised algebraic equations are solved using a point-implicit linear equation solver (ILU factorization scheme on a symmetric block Gauss-Seidel) in conjunction with an algebraic multi-grid (AMG) method to accelerate solution convergence.

Mesh independence analysis has been carried out considering two different meshes with increasing refinement for the same model and boundary conditions. In particular, the mesh has been locally refined in correspondence of the boundary layer to a value of $1 \cdot 10^{-5} m$, resulting in 1.7 million cells. The deviations between the computed maximum temperatures and plasma frequency corresponding to the initial and the refined mesh are less than 5%, showing good convergence for the first mesh, described at the beginning of the section. Therefore, the numerical model based on the first mesh has been considered a good compromise between accuracy and computational cost for all simulations. Similar results have been obtained for the two-dimensional cases where only the symmetry plane is considered.

Table 5

Boundary conditions used for CFD simulations. For each altitude case, standard values of asymptotic temperature and pressure are selected. Mach number is varied to cover different flight conditions. Turbulence model, wall boundary conditions, and inlet species concentration are common to all cases.

Altitude [km]	Temperature [K]	Pressure [Pa]	Mach number	Velocity [m/s]	Reynolds number
30	226	1171	5.7	1727	$7.85 \cdot 10^6$
			10.0	3022	$1.37 \cdot 10^7$
			13.3	4025	$1.83 \cdot 10^7$
			16.6	5034	$2.29 \cdot 10^7$
			18.6	5637	$2.56 \cdot 10^7$
40	251	267	10.4	3189	$2.71 \cdot 10^6$
			13.2	4210	$3.58 \cdot 10^6$
			16.0	5103	$4.34 \cdot 10^6$
			18.6	5933	$5.05 \cdot 10^6$
			13.2	4362	$9.09 \cdot 10^5$
50	270	75	16.0	5287	$1.10 \cdot 10^6$
			18.6	6147	$1.28 \cdot 10^6$
			9.6	3010	$1.97 \cdot 10^5$
60	245	20	12.7	4016	$2.60 \cdot 10^5$
			15.9	5018	$3.20 \cdot 10^5$
			18.6	5863	$3.83 \cdot 10^5$

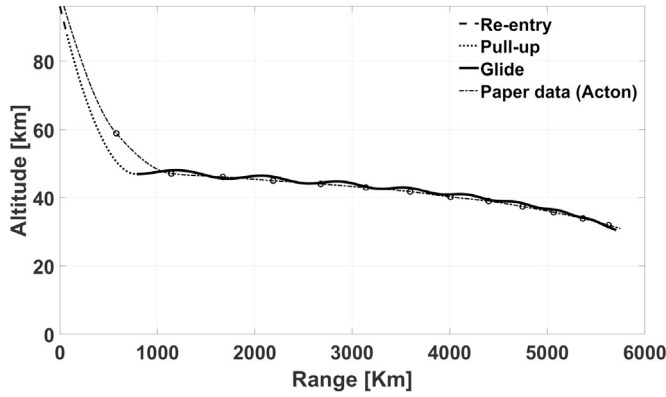


Fig. 7. Modelled trajectory used to define flight conditions for CFD analysis. Re-entry, pull-up and glide phases are compared with J.M. Acton's results [19] for a planar trajectory.

4. Results

4.1. Trajectory simulation results

Altitude and Mach number variations during the glide phase are shown in Figs. 7 and 8. A lift-to-drag ratio of $L/D = 3.3$ in the gliding phase is able to reproduce to the planned HTV-2 trajectory. A lift coefficient of $C_L = 0.1$ is selected to damp oscillations. A comparison between different constant bank manoeuvres is carried out to evaluate the effects on the cross-range, as reported in Fig. 9a. Fig. 9b shows how high bank angles impact the achieved distance. The distance from the starting point of the gliding to the end of the manoeuvre decreases with the second power of the bank angle. In Fig. 9c and Fig. 9d it is evident that higher bank angles result in rapid altitude loss and high speed during the descent. Moreover, without active control, oscillations in altitude and velocity occur if constant C_L is selected. High speed at low altitudes produces a very intense lift force, which overshoots the equilibrium value and results in a skip-glide trajectory with a continuous exchange of kinetic and potential energy. In addition, the effect of C_D and C_L variation is analysed in Fig. 10, where the efficiency and the lift coefficient have been scaled proportionally. Some results are interpolated to avoid excessive oscillations. Higher efficiencies allow longer flight paths; low-lift bodies achieve very high speeds during the last phase of the flight.

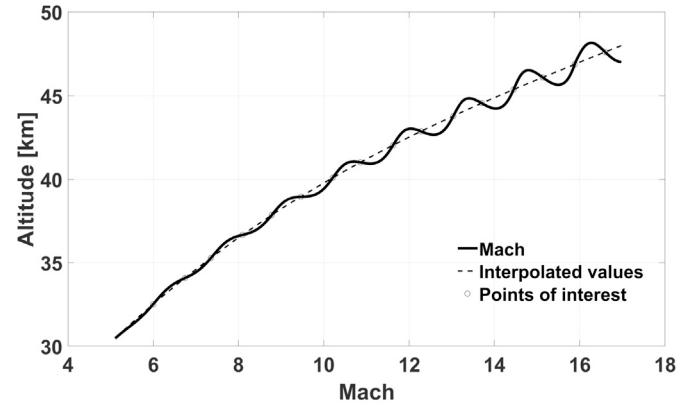


Fig. 8. Altitude and Mach number values during the glide phase. The oscillating behaviour can be explained by considering a constant C_L assumption, which causes a phugoid path.

4.2. Two-dimensional CFD simulation results

Parameters of interest on the upper and lower surfaces of the two-dimensional body are shown in Fig. 11, relative to a CFD simulation carried out at $M = 10$ and $h = 40$ km. A very high temperature is reached at the nose of the body, with a heat flux in the order of -10^6 W/m^2 . On the lower surface, a constant temperature of 1400K is reached, thus it is characterised by higher ablation of material. The lifting capability of the body is demonstrated since the pressure on the lower surface is higher than the pressure on the back. Moreover, the higher the temperature, the higher the density of electrons. This is due to more intense dissociation reactions, which result in the ionisation of chemical species. Consequently, the plasma frequency, given by equation (20), has a peak at the stagnation point. Fig. 12 and Fig. 13 show some analytical trends of how temperature, heat flux, density of electrons, and plasma frequency vary in the nose region at altitudes of 30 km and 60 km. The results have been derived by interpolating the outcomes of various flight conditions listed in Table 5. Data from CFD simulations have been interpolated to evaluate the variation of temperature, heat flux, density of electrons, and plasma frequency as a function of Mach and altitude. The results for a point on the glider's lower surface are presented in Fig. 14.

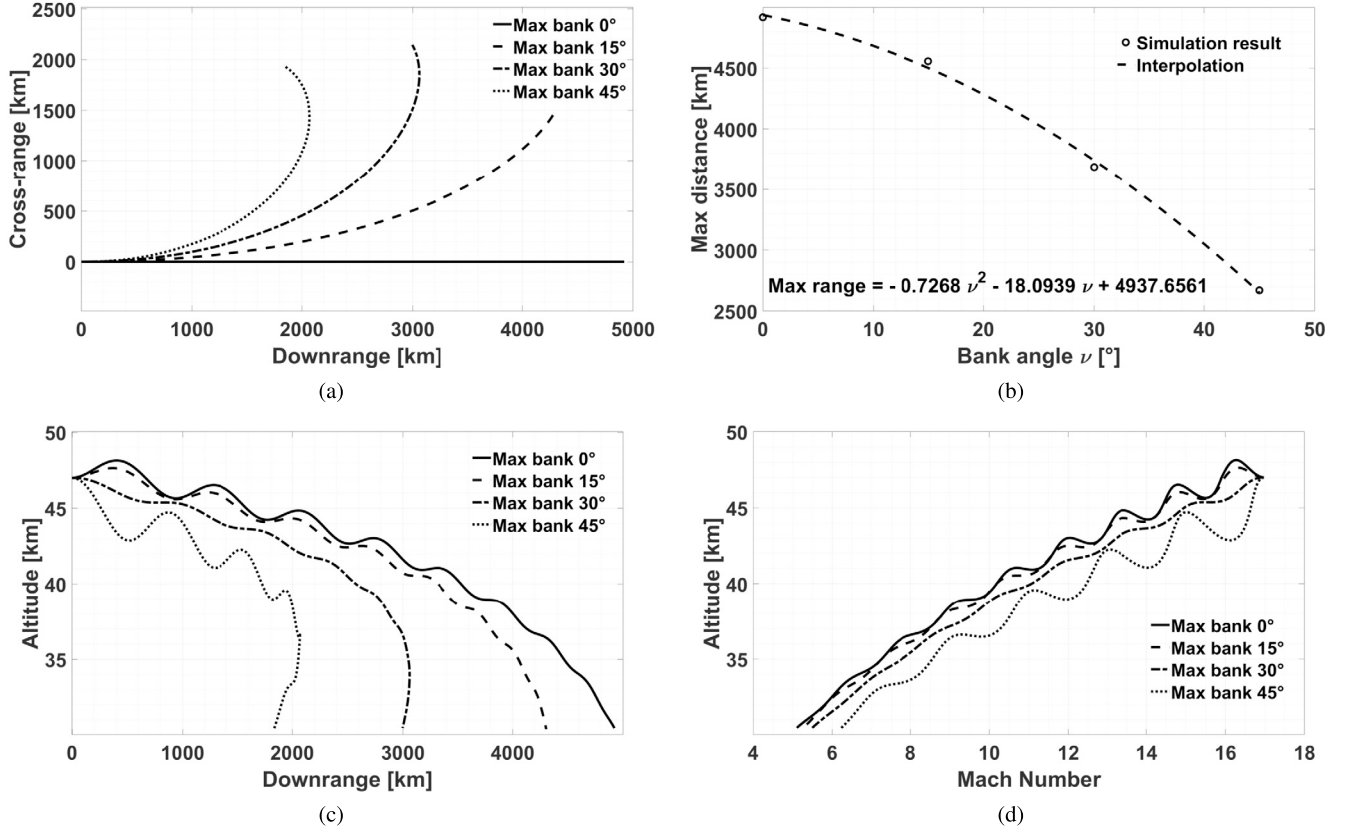


Fig. 9. Effects of the increasing variation of the bank angle on the glider's trajectory. (a) Cross-range and downrange results are shown: high bank angles allow operationally relevant cross-ranges, reducing the maximum downrange. (b) Interpolated values of final distance achievable as a function of bank angle; the distance reduces with the squared power of the bank angle. (c) High values of bank angles cause significant altitude loss reducing the downrange. (d) High values of bank angles increase the glider's speed during the descent.

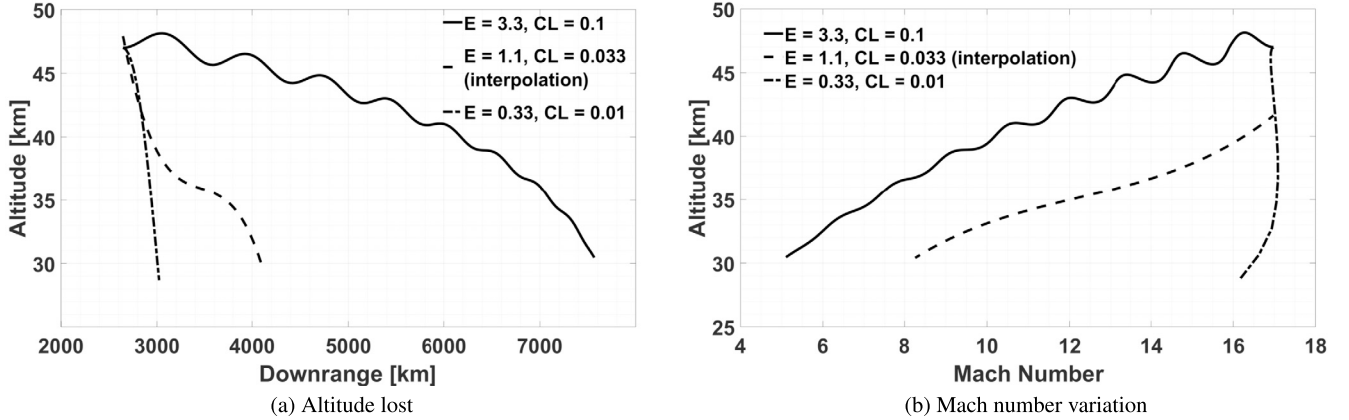


Fig. 10. Effects of lift-to-drag reduction. (a) Low L/D values cause a strong altitude variation in a shorter downrange. (b) Different values of L/D also impact the Mach number achieved during the gliding phase at each altitude.

4.3. Three-dimensional CFD simulation results

Due to the angle of attack $\alpha = 15^\circ$, the stagnation line on the nose is placed on the lower surface, and this creates a bow shock extending towards the back of the body. The resulting entropic region on the top part of the glider extends for tens of centimetres. Here, the combination of high temperature and low speed (reduced convective time) allows chemical reactions to complete, producing highly ionised gas. Therefore, free electrons are formed and dragged away from the airflow. As a result, a thick ionised wake is formed on the back of the body, where

the low density of the gas mixture does not allow the recombination of dissociated particles, as shown in Fig. 15. Regarding Fig. 16, high plasma frequency up to 1 GHz is reached only in the nose region. High values also characterise the thin layer of plasma that covers the lower surface. The different collision frequency contributions are evaluated with equations (22), (24) and (25). Contours are shown in Figs. 17, 18 and 19 respectively.

The collision frequency between electrons and all other species in the fluid is evaluated using equation (21). It is valid only in the plasma region in Fig. 15, where the density of electrons is high enough that

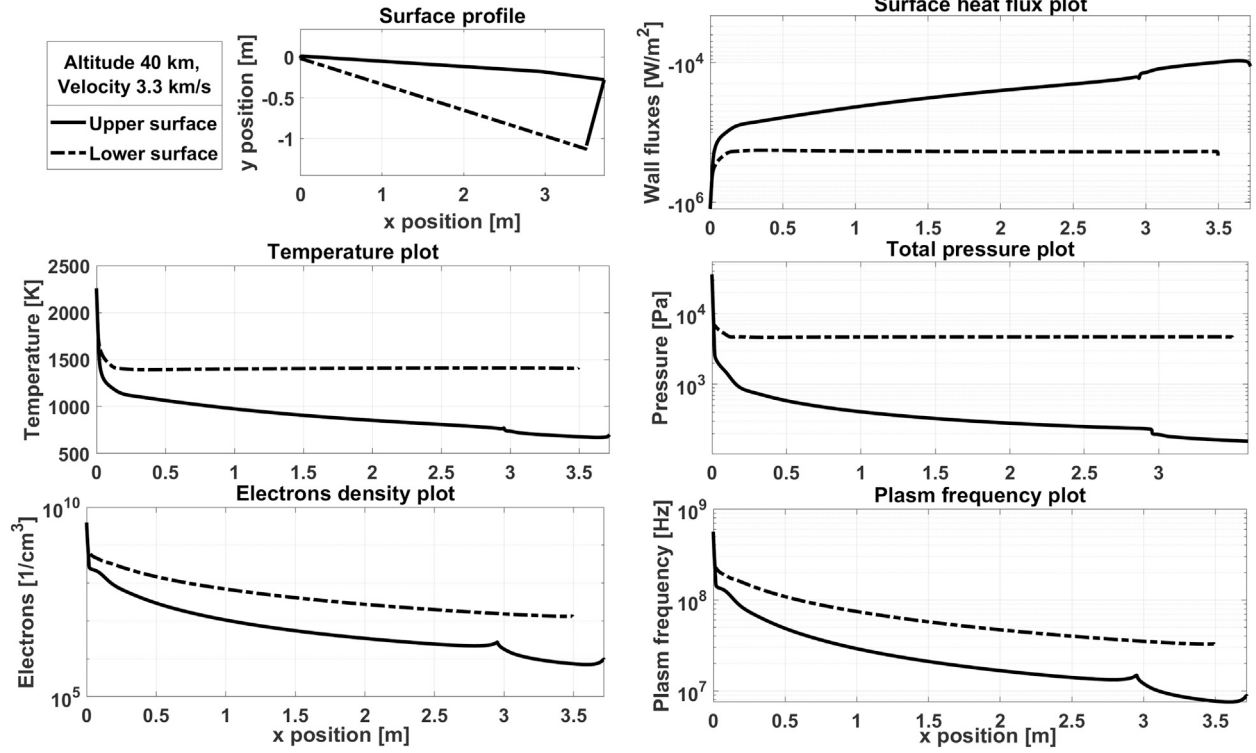


Fig. 11. Results of the bi-dimensional CFD simulation performed for $h = 40$ km and $M = 10$. Parameters of interest are presented for the lower surface (dotted line) and the upper surface. High temperatures are achieved at the nose location. Thus, the density of electrons is higher around the nose zone. In addition, the temperature on the lower face is constant and equal to 1400 K.

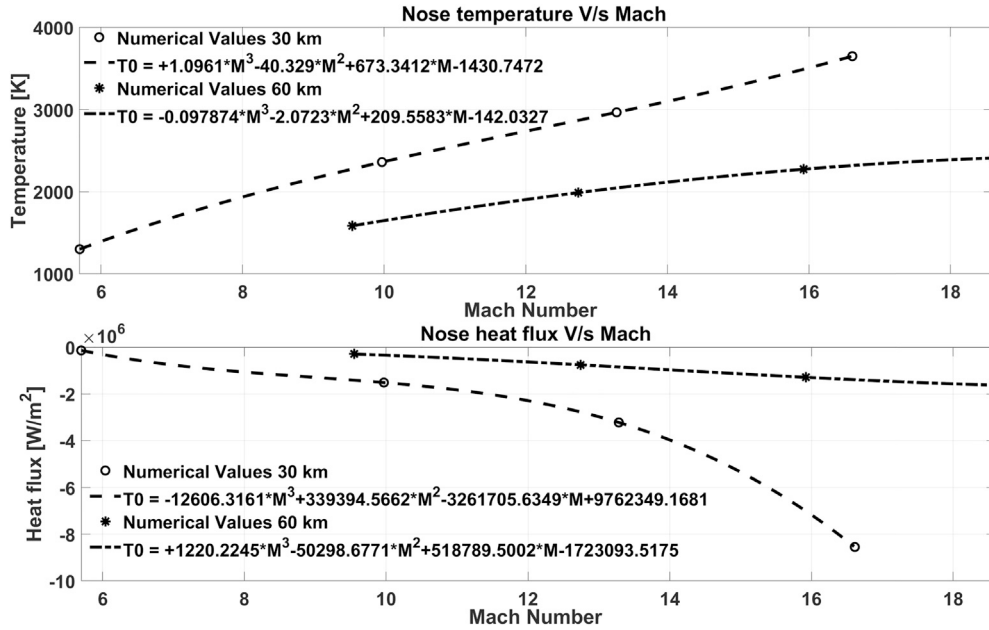


Fig. 12. Results of temperature and heat flux at the stagnation point for two altitudes ($h = 60$ km and $h = 30$ km). For both cases, interpolation of results is shown: outcomes of flight conditions presented in Table 5 have been used. Higher temperatures and heat fluxes are achieved at lower altitudes.

the gas could be considered plasma. Basically, in the assumed flight condition, electron-neutral collisions make up the majority of the total collision frequency. Values of plasma frequency, f_p , and collision frequency, ν_c , at a distance d along the normal to the surface are plotted

in Fig. 20 and Fig. 21. Considering l as the distance from the nose tip, measured along the longitudinal axis of the body, segments normal to the surface are defined at $l = 0$ m, $l = 1$ m, and $l = 2$ m, on both the upper and lower surfaces and the leading edge (l.e.).

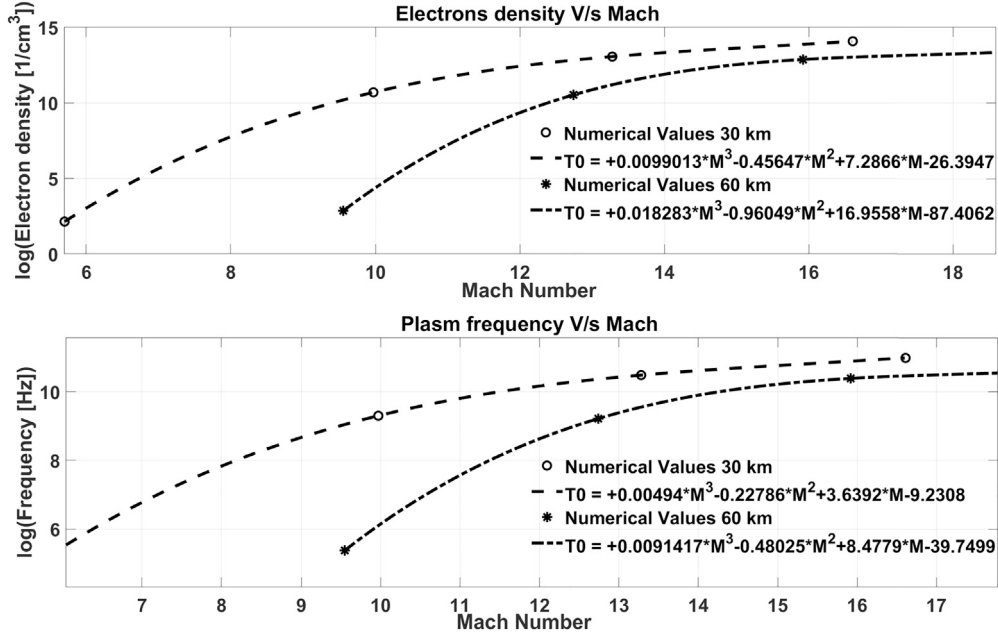


Fig. 13. Results of the density of electrons and plasma frequency at the stagnation point for two altitudes ($h = 30$ km and $h = 60$ km). Electron density increases with air density due to a higher number of molecules per volume unit that dissociates. Parabolic laws of variation of these parameters are obtained by interpolating the output of the simulations.

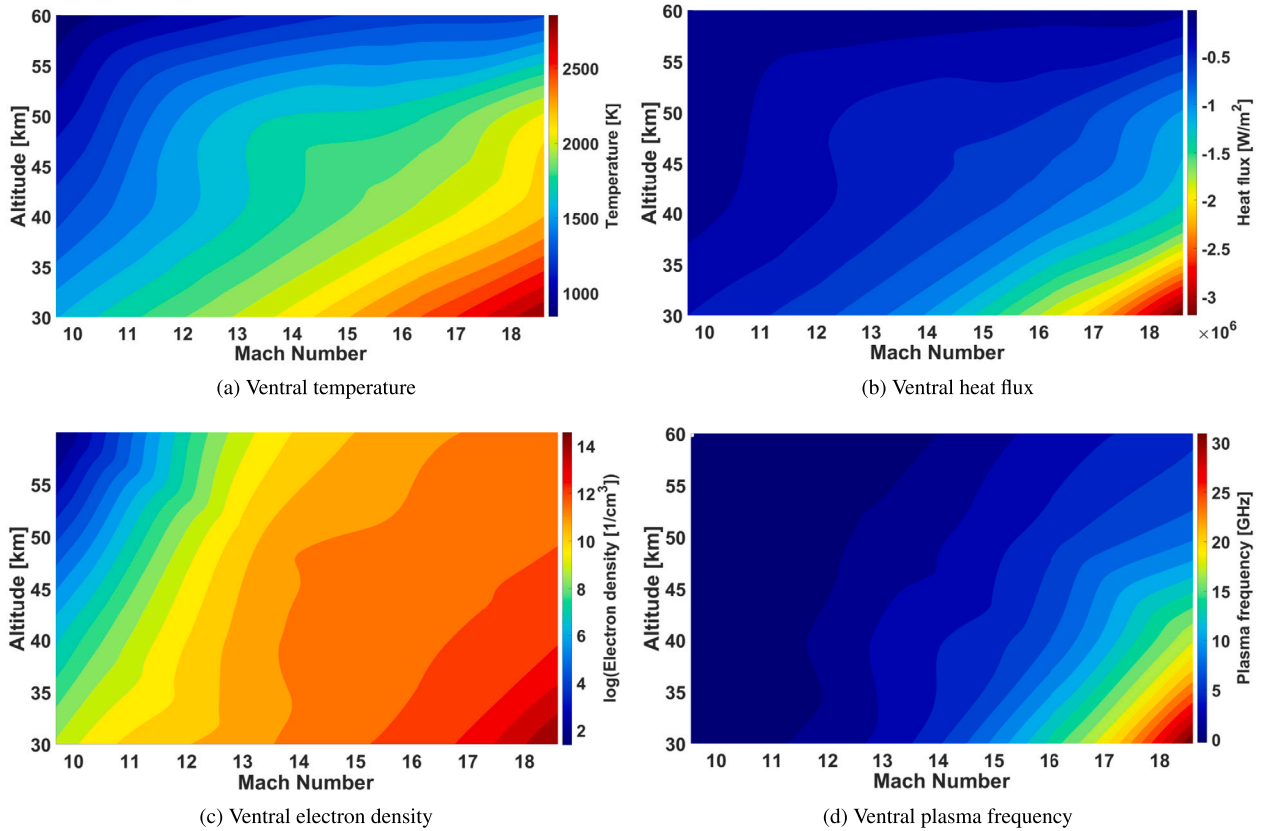


Fig. 14. Contours of (a) temperature, (b) heat flux, (c) electron density and (d) plasma frequency on the glider's lower surface. Shown results are obtained in the longitudinal symmetry plane corresponding to the point whose projection on the longitudinal axis is equal to $l = 0.5$ m from the nose tip. Variations of these parameters are shown as a function of Mach number and altitude.

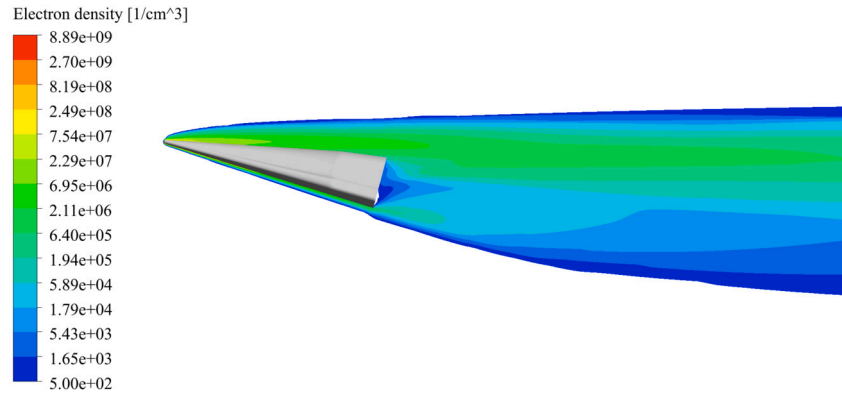


Fig. 15. Electron density contours for the three-dimensional CFD analysis for the gliding phase at $M = 10$ and $h = 40$ km. In these conditions, an ionised wake is formed on the back of the body, where, due to the low density of the gas mixture, there is no recombination of dissociated particles.

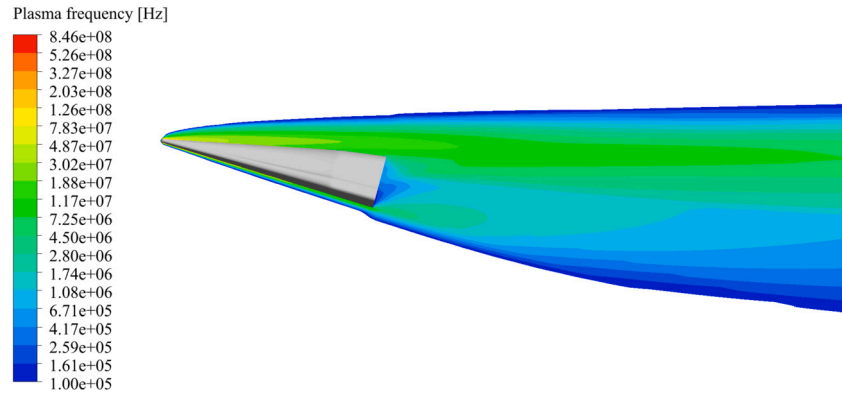


Fig. 16. Plasma frequency contours for the three-dimensional CFD analysis for the gliding phase at $M = 10$ and $h = 40$ km. Values of about 1 GHz are reached only close to the nose region.

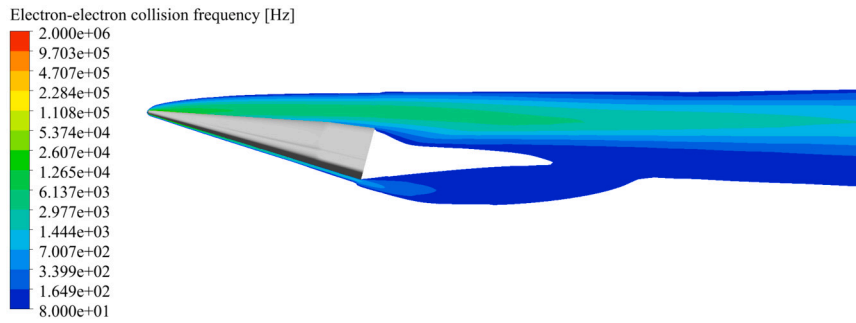


Fig. 17. Electron-electron collision contours in the symmetry plane for the three-dimensional CFD analysis for the gliding phase at $M = 10$ and $h = 40$ km.

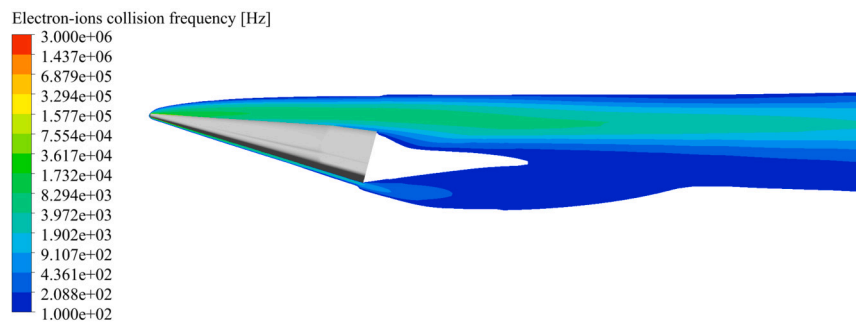


Fig. 18. Electron-ion collision contours in the symmetry plane for the three-dimensional CFD analysis for the gliding phase at $M = 10$ and $h = 40$ km.

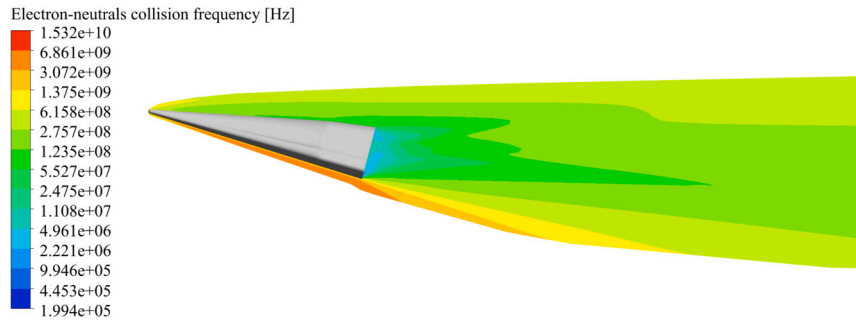


Fig. 19. Electron-neutral collision frequency contours in the symmetry plane for the three-dimensional CFD analysis of the gliding phase at $M = 10$ and $h = 40$ km.

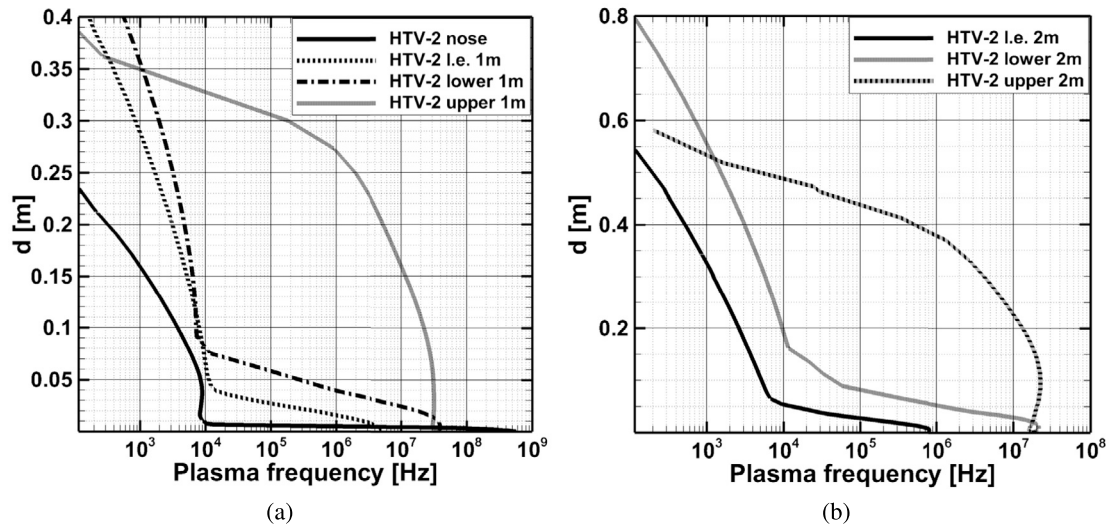


Fig. 20. Values of plasma frequency for different distances d along the normal to the surface. Three cases are shown, representing the distance l from the nose tip, measured along the longitudinal axis of the body: (a) $l = 0$ m, $l = 1$ m, (b) $l = 2$ m. These results are computed both for the lower and upper surface and the leading edge (l.e.). Peak values around 1 GHz are reached in a thin layer around the nose tip.

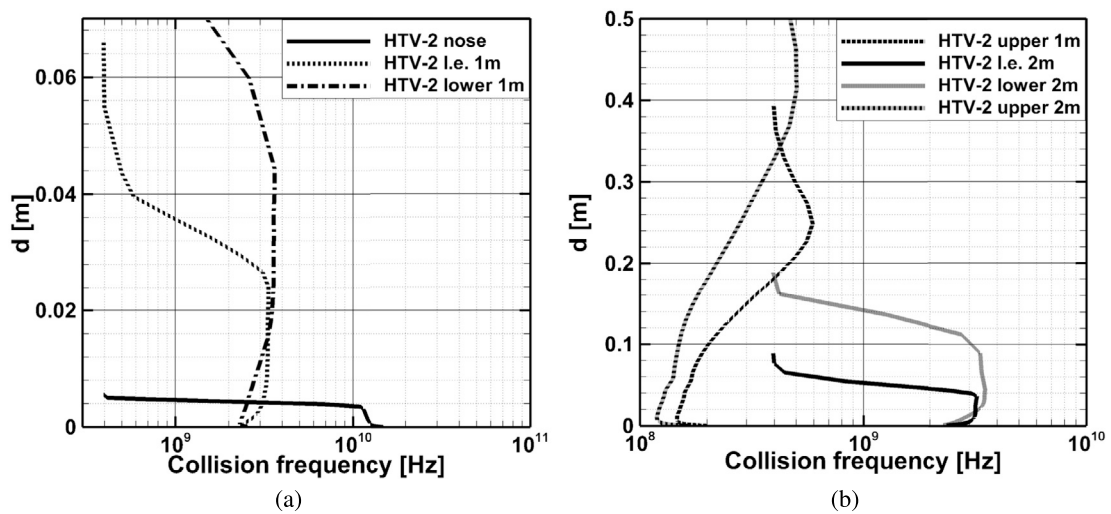


Fig. 21. Values of collision frequency for different distances d along the normal to the surface. Three cases are shown, representing the distance l from the nose tip, measured along the longitudinal axis of the body: (a) $l = 0$ m, $l = 1$ m, (b) $l = 2$ m. These results are computed both for the lower and upper surface and the leading edge (l.e.). Values peak around 10 GHz close to the nose tip.

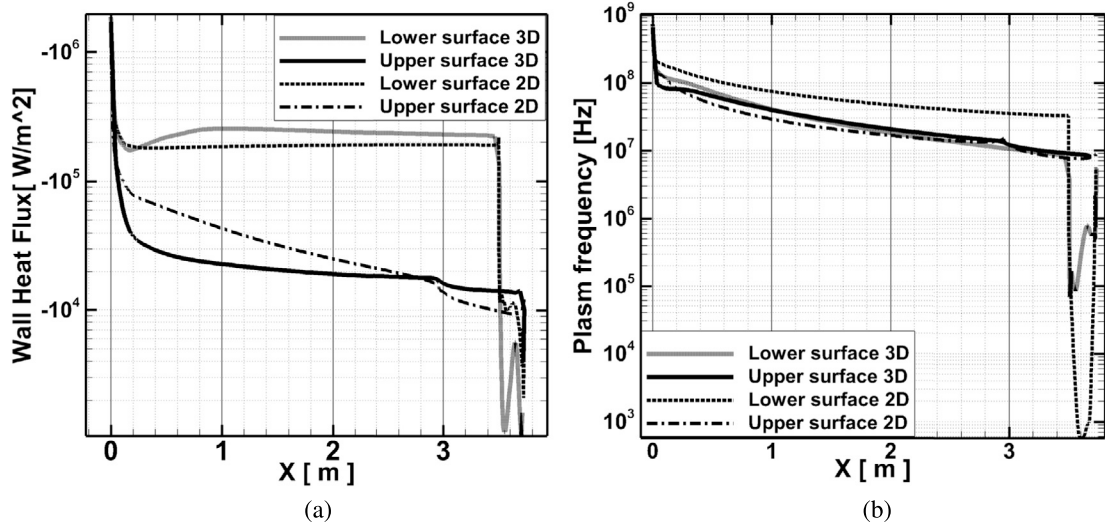


Fig. 22. Comparison of two-dimensional and three-dimensional CFD results on the longitudinal profile of the body. (a) 2D simulations underestimate the lower surface's heat flux. (b) Regarding plasma frequency, it is visible that 2D simulations of the upper surface give a good approximation for the more complex 3D geometry.

5. Discussion

Results on the longitudinal profile of the body have been compared between the 2D and 3D simulations, as shown in Fig. 22. Heat fluxes on the lower surface are underestimated in 2D simulations. This can be explained considering that the two-dimensional geometry behaves similarly to a cylinder under the effect of a transversal flow, while the real geometry can be related to a sphere. It is known that the heat flux is inversely proportional to the stand-off distance of the shock wave, which is lower for the three-dimensional body. The plasma frequency on the surface of the 3D geometry is compared with the two-dimensional case. It can be noticed that the values on the top and bottom surfaces of the 3D case are practically identical and similar to the values on the upper surface of the 2D study. This can be explained considering that even if higher temperatures—thus ionisation—characterise the lower surface in the 3D case, the effect of the cross-flow reduces the concentration of free electrons in the symmetry plane. For this reason, values on the upper surface of the 2D body could give a good approximation of the more complex geometry. In the 3D geometry results shown in Fig. 20, the plasma frequency reaches a peak of about 1 GHz in a thin layer around the nose tip, where the collision frequency is around 10 GHz. Comparing the profiles at a distance $l = 1\text{ m}$ from the nose, it is evident that high frequencies characterise the fluid on the lower surface. Here, the plasma sheath is about 8 cm thick, while on the back surface, the ionised region with plasma frequencies in the order of tens of MHz extends up to a distance of 30 cm from the surface. As a general trend, moving backwards, both plasma and collision frequencies tend to reduce, while the ionised region grows in thickness. Finally, it has been verified that it is possible to conduct simulations on a simplified geometry, shown in Fig. 23, to study the field around a real glider using the same boundary conditions of the three-dimensional case, and imposing a laminar boundary layer condition to further reduce the computational cost. A simplified geometry case was analysed to compare the results since it can allow shorter simulation timings thanks to the coarser mesh employed (165,000 hexahedral cells). In Fig. 24 it is reported how plasma frequency changes with distance from the surface. The results for the conic shape are very similar to those for the complex HTV-2 shape.

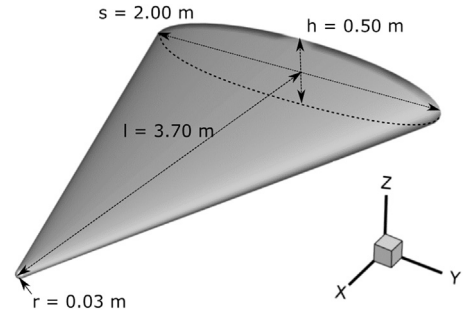


Fig. 23. Geometric features of the simplified glider body used for CFD simulations. A conic-shaped body has been modelled based on dimensions of the real HTV-2 glider geometry.

6. Conclusions

This work aimed to provide valuable information about the aero-thermo-chemical behaviour of the fluid in the region surrounding a hypersonic glider. A flight mechanics simulation tool has been developed to define the input flight conditions for computational fluid dynamics (CFD) simulations. The analyses of the ascent, re-entry, and glide phases shed light on the impact of different factors, such as launch conditions, aerodynamic parameters, and manoeuvres, on the overall trajectory of the glider. Additionally, they allow to identify potential ranges of the hypersonic gliders. A detailed CFD analysis has been performed to characterise the fluid field surrounding the hypersonic glider. By employing advanced numerical techniques, the heat transfer on the surface is analysed. Moreover, important plasma parameters are determined close to the body. These variables, such as plasma frequency and collision frequency, are essential to characterising electromagnetic interference in wave propagation. A gas mixture potentially composed of 11 species is considered to allow the solution of chemical volumetric reactions and obtain the density of electrons where extreme temperatures cause the ionisation of chemical species. Among other results, it is highlighted that high plasma and collision frequencies, in the order of tens of GHz, are generated close to the nose of the body and to the lower surface. For this reason, it is necessary to use high-frequency waves to withstand electromagnetic interference while transmitting into the plasma sheath. Firstly, regarding the limiting assumptions of the study,

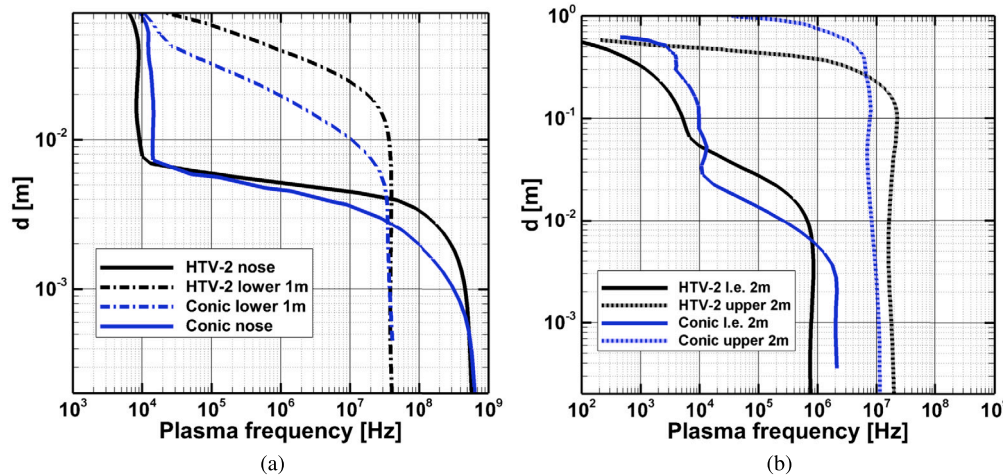


Fig. 24. Comparison of plasma frequency results for the cases of the simplified conic-shape geometry and the more complex HTV-2 one. Results are presented at sections (a) $l = 0$ m, $l = 1$ m, (b) $l = 2$ m. The simplified model can be employed as a reliable geometry to reduce computing time without reducing the reliability of the results.

the flight mechanics model solves point-mass dynamics without any attitude control logic. Secondly, the analyses do not consider the conjugate heat transfer between the body and the surrounding fluid. The insights gained represent an advance in the characterisation of the fluid field and support further research dealing with the datalink and radar detection of hypersonic systems. The determination of plasma parameters, not easily accessible in the literature, is essential to studying the radar cross-section of a hypersonic cutting-edge technology, which is a crucial aspect of boosting the Defence capability against new threats coming from space.

CRediT authorship contribution statement

Ragnoli Giona: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Savino Raffaele:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Cecere Anselmo:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis. **Rigamonti Marco:** Writing – review & editing, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used QuillBot AI in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding

This research did not receive any specific grants from funding agencies in the public, commercial, or not-for-profit sectors.

References

- [1] V. Theofilis, S. Pirozzoli, P. Martin, Special issue on the fluid mechanics of hypersonic flight, *Theor. Comput. Fluid Dyn.* 36 (1) (2022) 1–8.
- [2] J. Anderson, *Hypersonic and High-Temperature Gas Dynamics*, AIAA Education Series, American Institute of Aeronautics and Astronautics, 2006.
- [3] T. Tian Zhang, Z. Guo Wang, W. Huang, L. Yan, Parameterization and optimization of hypersonic-gliding vehicle configurations during conceptual design, *Aerosp. Sci. Technol.* 58 (2016) 225–234.
- [4] C. Pellegrini, M. Balat-Pichelin, O. Rapaud, E. Bêche, Y. Thébaud, Oxidation resistance up to 2600 K under air plasma of $(Zr/Hf)B_2$ composites containing 20 vol% AlN for hypersonic applications, *Ceram. Int.* 49 (19) (2023) 31634–31648.
- [5] J. Zhang, M. Li, B. Han, Analysis of electromagnetic waves reflected by re-entry plasma sheath based on cso-fdtd, *Phys. Scr.* 98 (08 2023), <https://doi.org/10.1088/1402-4896/acf0f9>.
- [6] G. Pezzella, G. Marino, G.C. Rufolo, Aerodynamic database development of the esa intermediate experimental vehicle, *Acta Astronaut.* 94 (1) (2014) 57–72.
- [7] K. Brockmann, D. Stefanovich, et al., *Hypersonic Boost-Glide Systems and Hypersonic Cruise Missiles: Challenges for the Missile Technology Control Regime*, SIPRI: Stockholm International Peace Research Institute, 2022.
- [8] D.M. Van Wie, *Hypersonics: past, present, and potential future*, Johns Hopkins APL Tech. Dig. 35 (4) (2021) 335–341.
- [9] K.M. Sayler, *Hypersonic weapons: Background and issues for congress*, Tech. rep., Congressional Research Service, February 2023.
- [10] G. Li, H. Zhang, G. Tang, Maneuver characteristics analysis for hypersonic glide vehicles, *Aerosp. Sci. Technol.* 43 (2015) 321–328.
- [11] G. Zuppari, R. Savino, G. Mongelluzzo, Aero-thermo-dynamic analysis of a low ballistic coefficient deployable capsule in Earth re-entry, *Acta Astronaut.* 127 (2016) 593–602, <https://doi.org/10.1016/j.actastro.2016.06.041>.
- [12] C. Shao, L. Nie, W. Chen, Analysis of weakly ionized ablation plasma flows for a hypersonic vehicle, *Aerosp. Sci. Technol.* 51 (2016) 151–161, <https://doi.org/10.1016/j.ast.2016.02.005>.
- [13] N. Qinglin, Y. Zhichao, C. Biao, D. Shikui, Infrared radiation characteristics of a hypersonic vehicle under time-varying angles of attack, *Chin. J. Aeronaut.* 32 (4) (2019) 861–874.
- [14] Z. Bian, J. Li, L. Guo, X. Luo, Analyzing the electromagnetic scattering characteristics of a hypersonic vehicle based on the inhomogeneity zonal medium model, *IEEE Trans. Antennas Propag.* 69 (2) (2020) 971–982.
- [15] H. Zhang, J. Li, Z. Bian, L. Guo, Study on dynamic rcs of hypersonic vehicle, in: *2021 International Applied Computational Electromagnetics, Society (ACES-China) Symposium, IEEE*, 2021, pp. 1–2.
- [16] S.-h. Liu, L.-x. Guo, W.-t. Pan, W. Chen, Y.-f. Xiao, Po calculation for reduction in radar cross section of hypersonic targets using ram, *Phys. Plasmas* 25 (6) (2018).
- [17] T. Huynh, J. Kriz, Final environmental assessment for hypersonic technology vehicle 2 flight tests, Tech. rep., Teledyne Solutions Inc., Huntsville AL, 2009.
- [18] The MathWorks Inc, Natick, Massachusetts, United States, MATLAB version: 9.13.0, release R2022b (2022).
- [19] J.M. Acton, Hypersonic boost-glide weapons, *Sci. Glob. Secur.* 23 (3) (2015) 191–219.
- [20] D. Wright, Research note to hypersonic boost-glide weapons by James m. acton: analysis of the boost phase of the htv-2 hypersonic glider tests, *Sci. Glob. Secur.* 23 (3) (2015) 220–229.
- [21] G. Li, H. Zhang, G. Tang, Maneuver characteristics analysis for hypersonic glide vehicles, *Aerosp. Sci. Technol.* 43 (2015) 321–328.

- [22] P. Zipfel, Modeling and Simulation of Aerospace Vehicle Dynamics, AIAA Education Series, American Institute of Aeronautics and Astronautics, 2000.
- [23] C.L. Tracy, D. Wright, Modeling the performance of hypersonic boost-glide missiles, *Sci. Glob. Secur.* 28 (3) (2020) 135–170.
- [24] ANSYS Inc, Ansys Fluent User's Guide, July 2022, release 2022 R2.
- [25] M.G. Dunn, S. Kang, Theoretical and experimental studies of reentry plasmas, Tech. rep., NASA, 1973.
- [26] J.S. Evans, C.J. Schexnayder Jr, P.W. Huber, Boundary-layer electron profiles for entry of a blunts slender body at high altitude, Tech. rep., 1973.
- [27] R. Gupta, A Review of Reaction Rates and Thermodynamic and Transport Properties for an 11-Species Air Model for Chemical and Thermal Nonequilibrium Calculations to 30000 K, Langley Research Center, 1990.
- [28] R. Savino, M. D'Elia, V. Carandente, Plasma effect on radiofrequency communications for lifting reentry vehicles, *J. Spacecr. Rockets* 52 (2015) 417–425.
- [29] R. Savino, D. Paterna, M. De Stefano Fumo, M. D'Elia, Plasma-radiofrequency interactions around atmospheric re-entry vehicles: modelling and arc-jet simulation, *Open Aerosp. Eng. J.* 3 (2010) 76–85.
- [30] A.S. Richardson, 2019 nrl plasma formulary, Tech. rep., US Naval Research Laboratory, 2019.
- [31] J. Sheffield, D.H. Froula, S.H. Glenzer, N.C. Luhmann, Plasma scattering of electromagnetic radiation, in: *Theory and Measurement Techniques*, second edition, Academic Press Is an Imprint of Elsevier, 2011.
- [32] J.J. Bertin, Hypersonic Aerothermodynamics, AIAA Education Series, American Institute of Aeronautics and Astronautics, 1994.